Phase 2 Enhancement: Electrical Health Report & Consumption Intelligence

Project Context

The website already contains:

Existing Features

- Appliance-based load calculation
- Total Connected Load (kW/W)
- Daily Consumption (kWh/day)
- Monthly Consumption (kWh/month)
- Estimated Electricity Bill
- Top Consumer Analysis
- PDF Export
- Smart Insights Section

Smart Insights Implemented

- Recommended MCB
- Recommended Cable Size
- Recommended Inverter
- Recommended Solar System

The website follows a modern mobile-first design system:

- Dark theme
- Amber/Yellow accent color
- Rounded cards
- Premium calculator experience
- Existing report summary page

Do NOT redesign the current layout.

Build upon the existing report experience.

Goal

Transform the report from:

"Electrical Calculator"

into

"Electrical Health Assessment"

Users should feel they are receiving a professional energy audit.

Feature 1: Electrical Health Score

Placement

Insert immediately below:

🧠 Smart Insights

and above:

Consumption Breakdown

UI Card

Create a dedicated premium card.

Example:

Electrical Health Score

87 / 100

Excellent

Score Categories

Generate score using weighted rules.

Load Utilization (30%)

Evaluate:

Current Load vs Configured Capacity

Rules:

< 50% = 20 points

50%–80% = 30 points

80%–95% = 20 points

95%+ = 10 points

Energy Efficiency (30%)

Determine:

Percentage contribution of top appliance.

Rules:

Top appliance < 40% = 30 points

40–60% = 20 points

60%+ = 10 points

Safety Readiness (20%)

Check:

MCB Recommendation Available

Cable Recommendation Available

Both Available = 20 points

One Missing = 10 points

None = 0 points

Renewable Readiness (20%)

Monthly Units:

< 200 = 10 points

200–500 = 15 points

500+ = 20 points

Score Result

Generate:

0–40 = Poor

41–60 = Fair

61–80 = Good

81–100 = Excellent

Display:

Green = Excellent

Yellow = Good

Orange = Fair

Red = Poor

Maintain current design language.

Feature 2: Consumption Breakdown Visualization

Goal

Help users understand where energy is consumed.

Current report only lists appliances.

Add visual intelligence.

Group Appliances Into Categories

Lighting

Fans & Cooling

Kitchen

Entertainment

Office

Industrial

Other

Visualization

Create a responsive chart card.

Preferred:

Donut Chart

Alternative:

Horizontal Percentage Bar Chart

Display:

Cooling 68%

Kitchen 15%

Lighting 8%

Entertainment 5%

Other 4%

Additional Information

Display:

Highest Consumption Category

Example:

Cooling consumes 68% of total energy usage.

Feature 3: Energy Saving Opportunities

Placement

Below Consumption Breakdown.

Objective

Automatically generate actionable recommendations.

Example:

Potential Monthly Savings

₹1,800/month

Logic

Identify:

Top Energy Consuming Appliance

Generate appliance-specific suggestions.

Air Conditioner

Suggestions:

Increase temperature by 1–2°C

Use timer mode

Clean filters monthly

Switch to inverter AC

Refrigerator

Suggestions:

Check door sealing

Avoid frequent opening

Maintain rear ventilation

Lighting

Suggestions:

Replace CFL with LED

Turn off unused lights

Use occupancy sensors

Water Heater

Suggestions:

Use timer scheduling

Lower temperature setting

Insulate hot water pipes

Output

Display:

Top Consumer

Potential Savings Estimate

3–5 Smart Recommendations

Feature 4: High Consumption Alerts

Placement

Below Energy Saving Opportunities

Example

High Consumption Devices

⚠ Air Conditioner (2T)

20.0 kWh/day

⚠ Refrigerator

7.2 kWh/day

Logic

Flag appliances consuming:

    15% of daily usage

Display:

Consumption

Reason for flagging

Feature 5: Enhanced PDF Report

Goal

Convert PDF into a professional electrical assessment report.

New PDF Structure

Page 1

Executive Summary

Load

Consumption

Bill

Health Score

Page 2

Smart Insights

MCB

Cable

Inverter

Solar

Page 3

Consumption Breakdown

Category Distribution

Charts

Analysis

Page 4

Energy Saving Opportunities

Recommendations

Potential Savings

High Consumption Devices

PDF Design Requirements

Maintain existing branding.

Maintain dark/yellow theme.

Preserve current summary sections.

Add new sections without removing existing information.

Ensure clean page breaks.

Professional report appearance.

Suitable for sharing with:

Electricians

Homeowners

Solar Installers

Energy Consultants

Technical Requirements

- Reuse existing calculation results.
- No backend changes.
- Fully client-side.
- Responsive on mobile and desktop.
- Follow existing component architecture.

- Keep calculations deterministic.
- Ensure PDF generation remains performant.
- Build reusable helper functions for:
- Health Score Engine
- Consumption Analyzer
- Savings Recommendation Engine

Success Criteria

A user should finish a calculation and immediately understand:

✓ Is my electrical setup healthy?

✓ What is consuming the most electricity?

✓ Where can I save money?

✓ What actions should I take next?

The final experience should feel like a professional energy assessment report rather than a traditional calculator.